Loops and trace tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Loops and trace tables before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Loops and trace tables trace and record the first step you can explain without help.

Use W2: Trace before running for **Loop**s and **trace table**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Loops and trace tables explanation against the supplied facts before

discussing it.

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Count iterations and a stopping point

List every value visited and each running total. Then predict the result for range(2,2).

```
total = 0
for value in range(2, 6):
    total = total + value
print(total)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the range

> Explain the range. Apply the supplied Loops and trace tables facts and keep your reasoning.

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace before running

> Trace before running. Apply the supplied Loops and trace tables facts and keep your reasoning.

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Repair the reset

> Repair the reset. Apply the supplied Loops and trace tables facts and keep your reasoning.

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Build a repeated addition independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build a repeated addition independently. Apply the supplied Loops and trace tables facts and keep your reasoning.

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Check a contract limitation

Check a contract limitation. Apply the supplied Loops and trace tables facts and keep your reasoning.

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

Connect Loops and trace tables to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled **loop**s and **trace table**s **evidence** sheet to Python readiness portfolio. Show how you can trace variable values before and after each **loop iteration**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
>
> _____
>
> _____
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> _____
>
> _____

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Loops and trace tables

> Accumulate a short range. A running total adds the integers visited by range(1,5). Explain the labelled step before checking the changed condition.

Accumulate a short range. A running total adds the integers visited by range(1,5).

Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/6

Loops and trace tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Visit the range

Values are 1,2,3,4; the stop value 5 is excluded.



2. Trace running total

Totals after each addition are 1,3,6,10.



3. Print after the loop

The final displayed value is 10.

CHANGE ONE THING

Replace range(1,5) with range(1,1).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Accumulate a short range. A running total adds the integers visited by range(1,5).

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final displayed value is 10.

1. Trace running total

2. Print after the **loop**

3. Visit the range

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Print after the **loop** The final displayed value is 10. A **loop** repeats an operation for each visited value.

A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace range(1,5) with range(1,1). Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. No values are visited and the final **output** is 0.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: No values are visited and the final **output** is 0. The **accumulator** starts at zero and an empty **loop** never changes it.

Repair a missing building block

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python’s grouping rules. Short practice: Trace the operation order in both expressions.

Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> |---|
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> | |
> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Loops and trace tables picture from its labels, then record a relevant limit.

01. Loops and trace tables: the case



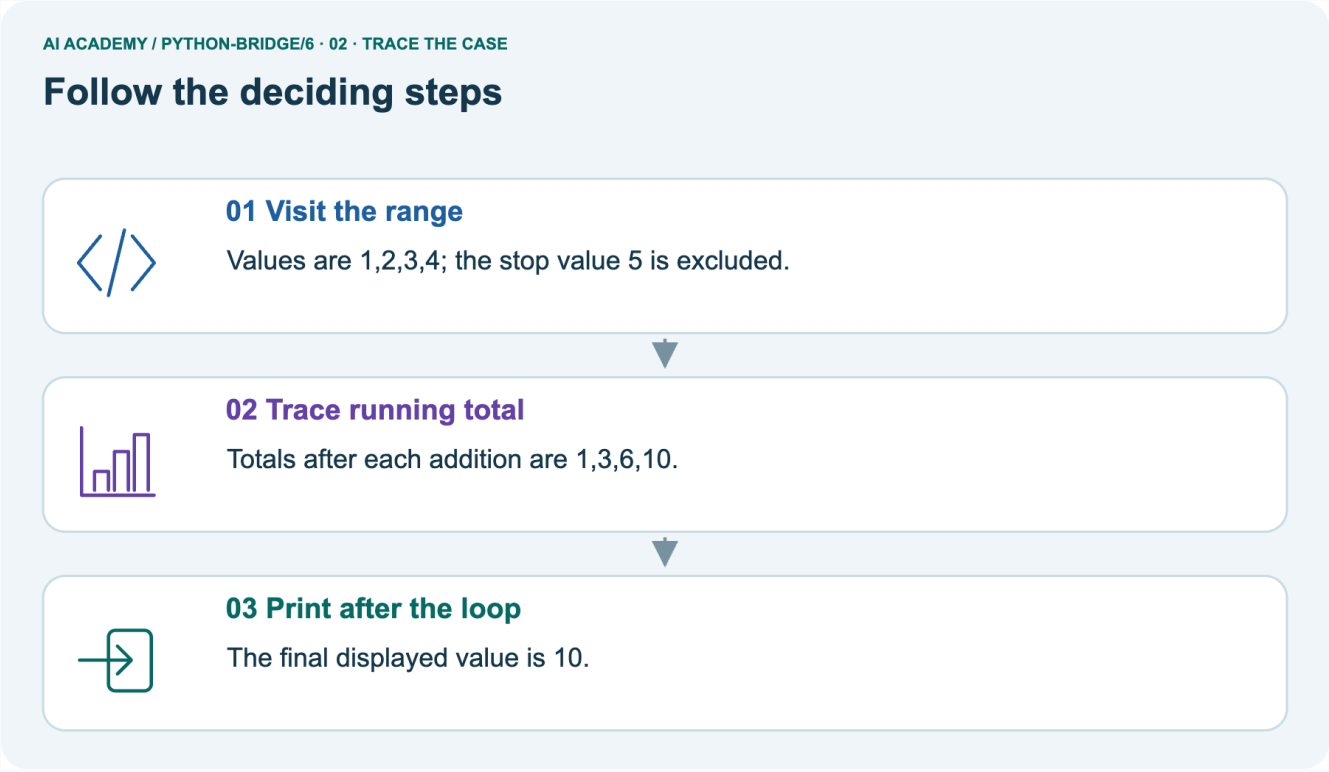
Follow the labelled picture with the matching explanation. Open the vector diagram

Accumulate a short range. A running total adds the integers visited by range(1,5). Supplied facts: Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



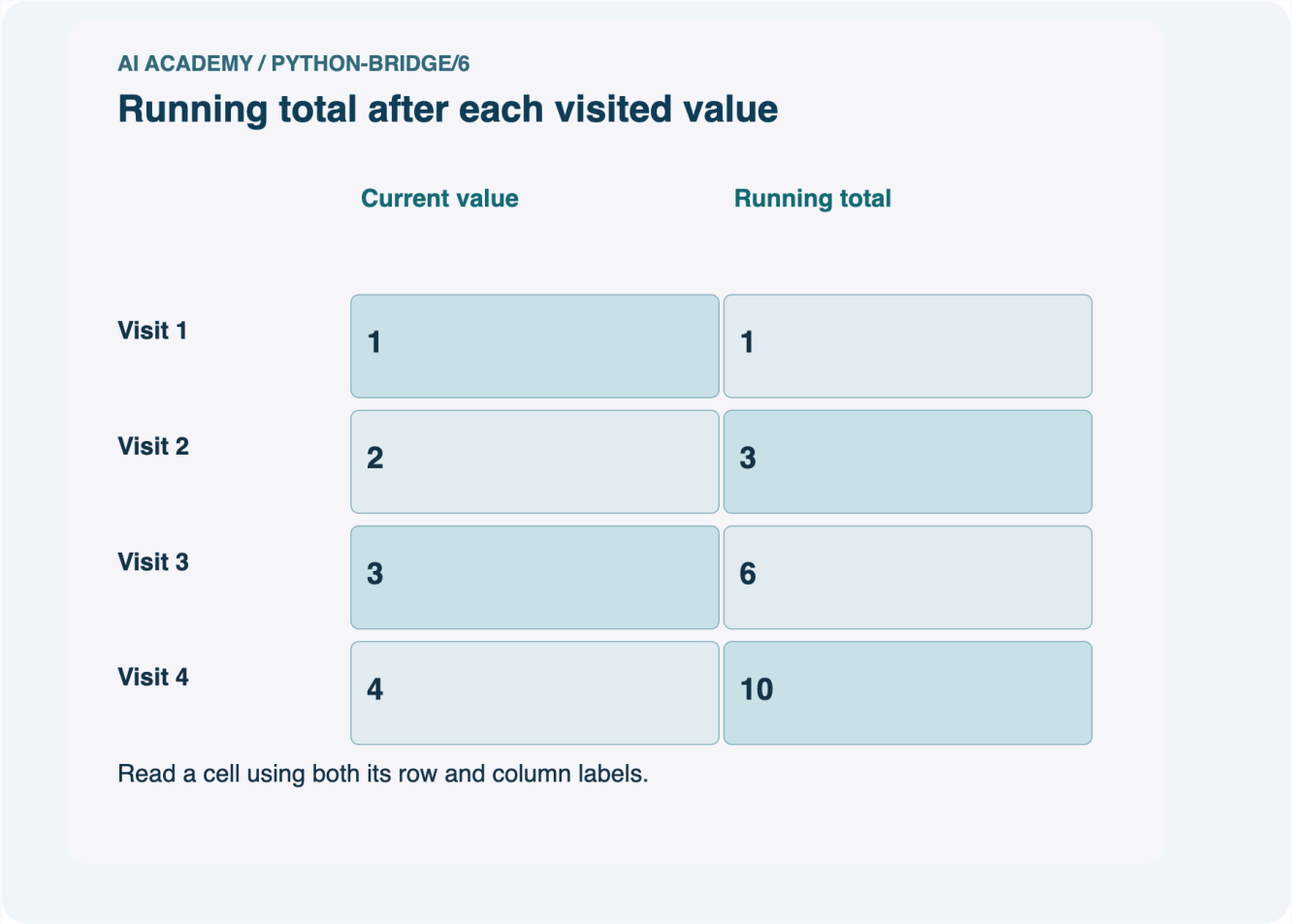
Follow the labelled picture with the matching explanation. Open the vector diagram

Visit the range: Values are 1,2,3,4; the stop value 5 is excluded. → Trace running total: Totals after each addition are 1,3,6,10. → Print after the **loop**: The final displayed value is 10.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Running total after each visited value



Follow the labelled picture with the matching explanation. Open the vector diagram

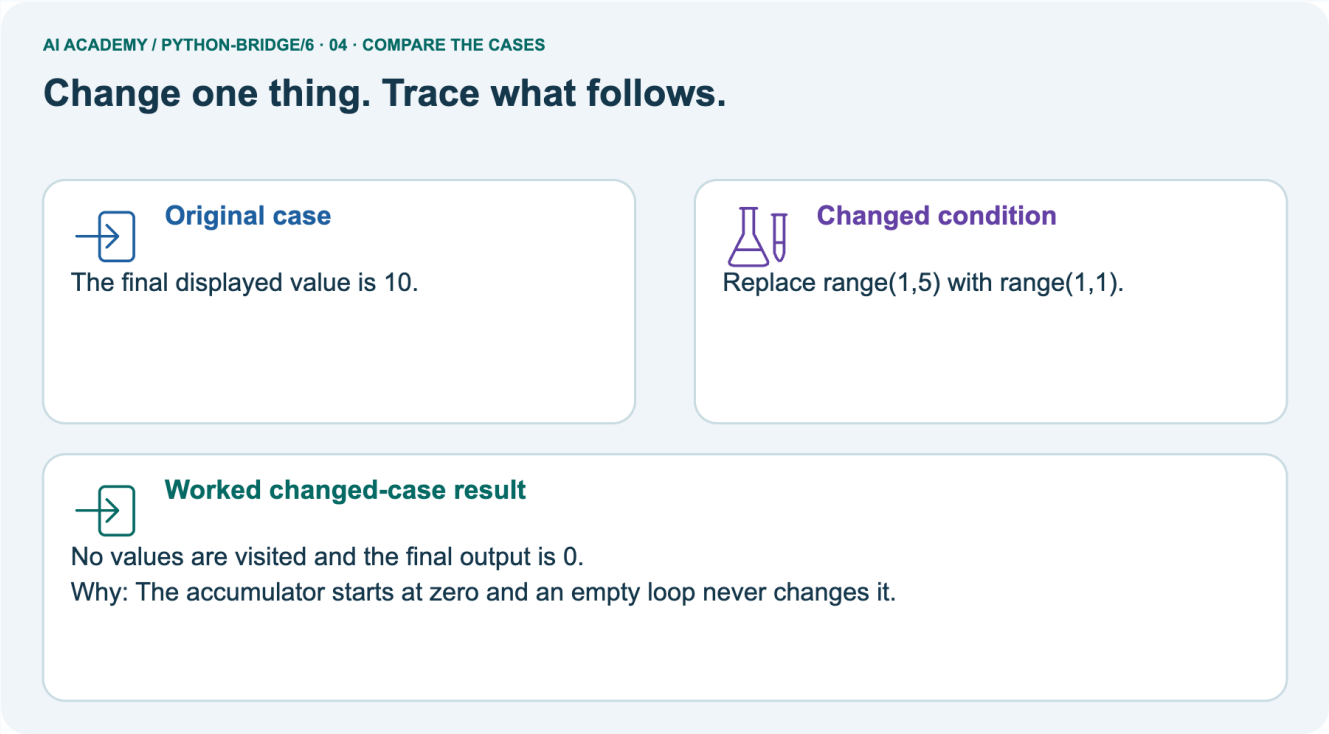
Running total after each visited value. Visit 1: Current value = 1, Running total = 1. Visit 2: Current value = 2, Running total = 3. Visit 3: Current value = 3, Running total = 6.

Visit 4: Current value = 4, Running total = 10.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

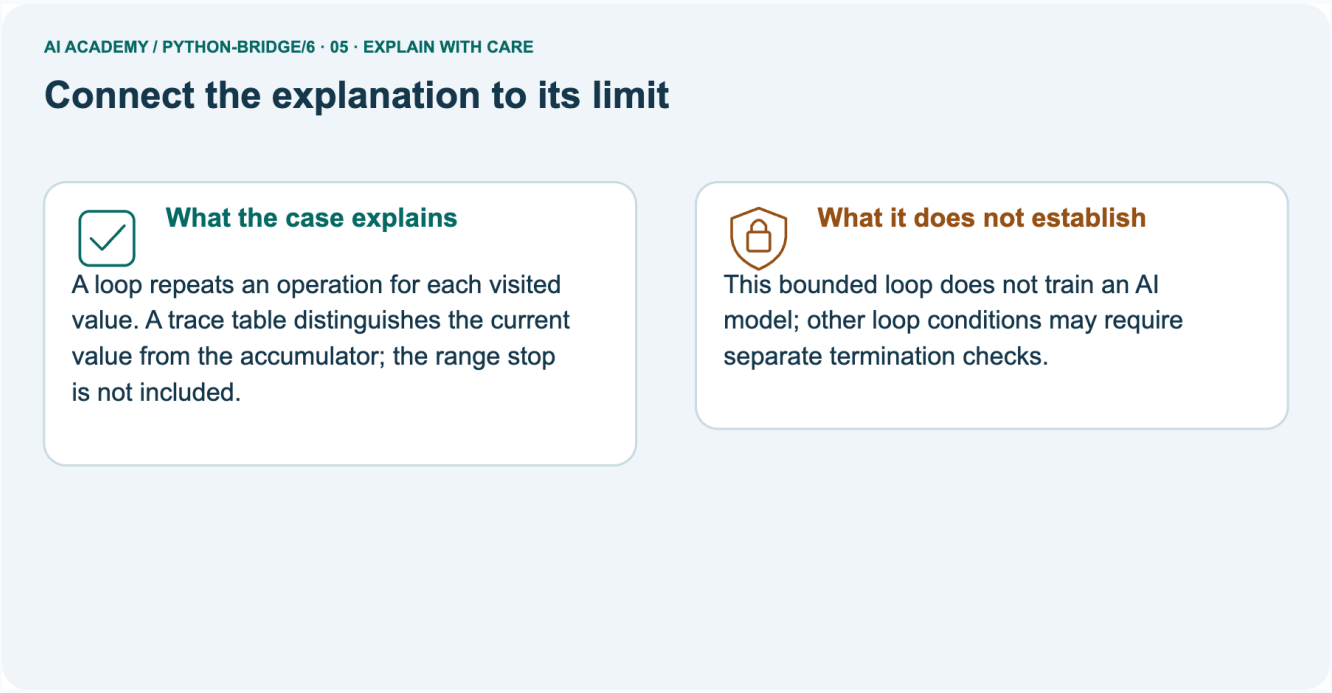
Original case: The final displayed value is 10. Changed condition: Replace range(1,5) with range(1,1). Expected result: No values are visited and the final **output** is 0.

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

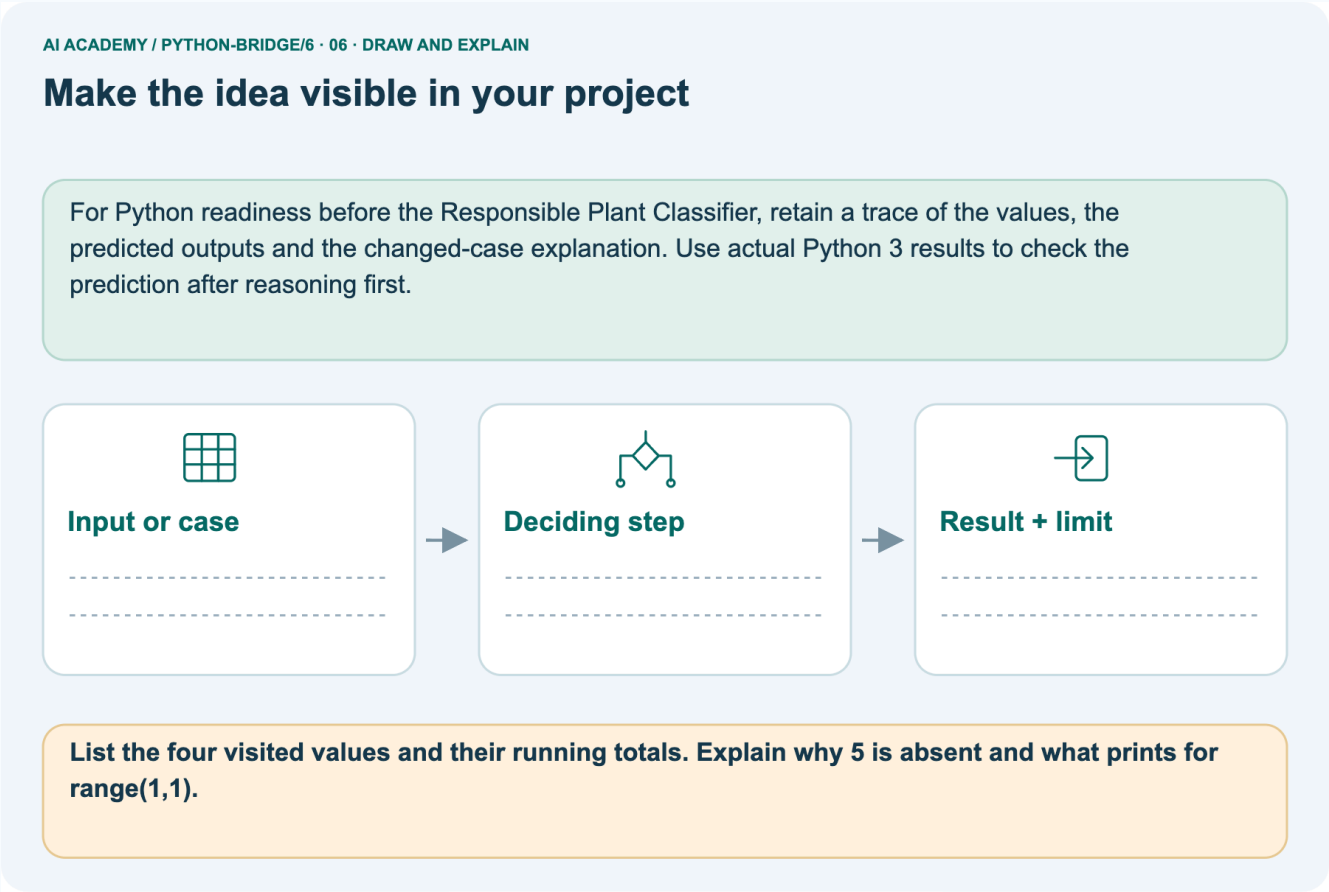
A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Limit: This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.